EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $g(x)$ be an antiderivative for $f(x)$. Then $\ell n\left(1+(g(x))^2\right)$ is an antiderivative for

(A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) none

MII001

2. Consider $f(x)=\frac{x^2}{1+x^3}$; $g(t)=\int f(t)dt$. If $g(1)=0$ then $g(x)$ equals –

(A) $\frac{1}{3}\ell n(1+x^3)$ (B) $\frac{1}{3}\ell n\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2}\ell n\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3}\ell n\left(\frac{1+x^3}{3}\right)$

MII002

3. $\int\frac{x^2(1-\ell n\,x)}{\ell n^4x-x^4}dx$ equals

(A) $\frac{1}{2}\ell n\left|\frac{x}{\ell n\,x}\right|-\frac{1}{4}\ell n|\ell n^2x-x^2|+C$ (B) $\frac{1}{4}\ell n\left|\frac{\ell nx-x}{\ell nx+x}\right|-\frac{1}{2}\tan^{-1}\left(\frac{\ell nx}{x}\right)+C$

(C) $\frac{1}{4}\ell n\left|\frac{\ell nx+x}{\ell nx-x}\right|+\frac{1}{2}\tan^{-1}\left(\frac{\ell nx}{x}\right)+C$ (D) $\frac{1}{4}\left(\ell n\left|\frac{\ell nx-x}{\ell nx+x}\right|+\tan^{-1}\left(\frac{\ell nx}{x}\right)\right)+C$

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MII003

4. Primitive of $\frac{3x^4-1}{(x^4+x+1)^2}$ w.r.t. x is -

(A) $\frac{x}{x^4+x+1}+c$ (B) $-\frac{x}{x^4+x+1}+c$ (C) $\frac{x+1}{x^4+x+1}+c$ (D) $-\frac{x+1}{x^4+x+1}+c$

MII004

5. $\int\frac{x^2+x}{(e^x+x+1)^2}dx$ equals

(A) $C-\ell n(1+(x+1)e^x)-\frac{1}{1+(x+1)e^x}$ (B) $C-\ell n(1+(x+1)e^x)+\frac{1}{1+(x-1)e^x}$

(C) $C-\ell n(1-(x^2-1)e^{-x})+\frac{1}{1+(x+1)e^x}$ (D) $C-\ell n(1+(x+1)e^{-x})-\frac{1}{1+(x+1)e^{-x}}$

MII005

6. $\int\frac{1-x^7}{x(1+x^7)}dx$ equals -

(A) $ln\,x+\frac{2}{7}\ln(1+x^7)+c$ (B) $ln\,x-\frac{2}{7}\ln(1-x^7)+c$

(C) $ln\,x-\frac{2}{7}\ln(1+x^7)+c$ (D) $ln\,x+\frac{2}{7}\ln(1-x^7)+c$

MII006

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7. Integral of $\sqrt{1+2\cot x(\cot x+\operatorname{cosec} x)}$ w.r.t. x is

(A) $2\ell n\cos\frac{x}{2}+c$ (B) $2\ell n\sin\frac{x}{2}+c$

(C) $\frac{1}{2}\ell n\cos\frac{x}{2}+c$ (D) $ln\ \sin x - ln(\operatorname{cosec} x - \cot x) + c$

MII007

8. Let $f(x)=\frac{2\sin^2 x-1}{\cos x}+\frac{\cos x(2\sin x+1)}{1+\sin x}$ then $\int e^x(f(x)+f'(x))dx$

(where c is the constant of integration)

(A) $e^x \tan x + c$ (B) $e^x \cot x + c$ (C) $e^x \operatorname{cosec}^2 x + c$ (D) $e^x \sec^2 x + c$

MII008

9. $\int e^x\left(\frac{x^2-3}{(x-1)^2}\right)dx$ is equal to –(where C is constant of integration)

(A) $e^x\frac{(x+3)}{(x-1)}+C$ (B) $e^x\left(\frac{x-3}{x-1}\right)+C$ (C) $e^x\left(\frac{x+1}{x-1}\right)+C$ (D) $e^x\left(\frac{1}{x-1}\right)^2+C$

MII009

10. $\int x.\frac{\ln\left(x+\sqrt{1+x^2}\right)}{\sqrt{1+x^2}}dx$ equals -

(A) $\sqrt{1+x^2}\ ln\left(x+\sqrt{1+x^2}\right)-x\ +c$ (B) $\frac{x}{2}.\left(x+\sqrt{1+x^2}\right)ln^2\ -\frac{x}{\sqrt{1+x^2}}+c$

(C) $\frac{x}{2}.ln^2\ \left(x+\sqrt{1+x^2}\right)+\frac{x}{\sqrt{1+x^2}}+c$ (D) $\sqrt{1+x^2}\ ln\left(x+\sqrt{1+x^2}\right)+x+c$

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MII010

11. The evaluation of $\int\frac{p\,x^{p+2q-1}-q\,x^{q-1}}{x^{2p+2q}+2x^{p+q}+1}dx$ is

(A) $-\frac{x^p}{x^{p+q}+1}+C$ (B) $\frac{x^q}{x^{p+q}+1}+C$ (C) $-\frac{x^q}{x^{p+q}+1}+C$ (D) $\frac{x^p}{x^{p+q}+1}+C$

MII011

12. The integral $\int\sqrt{\cot x}\,e^{\sqrt{\sin x}}\sqrt{\cos x}\,dx$ equals

(A) $\frac{\sqrt{\tan x}\,e^{\sqrt{\sin x}}}{\sqrt{\cos x}}+C$ (B) $2e^{\sqrt{\sin x}}+C$ (C) $-\frac{1}{2}e^{\sqrt{\sin x}}+C$ (D) $\frac{\sqrt{\cot x}\,e^{\sqrt{\sin x}}}{2\sqrt{\cos x}}+C$

MII012

13. $\int(\sin(101x)\cdot\sin^{99}x)dx$ equals

(A) $\frac{\sin(100x)(\sin x)^{100}}{100}+C$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100}+C$

(C) $\frac{\cos(100x)(\cos x)^{100}}{100}+C$ (D) $\frac{\sin(100x)(\sin x)^{101}}{101}+C$

MII013

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14. $\int \sec^2\theta(\sec\theta+\tan\theta)^2 d\theta$

(A) $\frac{(\sec\theta+\tan\theta)}{2}[2+\tan\theta(\sec\theta+\tan\theta)]+C$

(B) $\frac{(\sec\theta+\tan\theta)}{3}[2+4\tan\theta(\sec\theta+\tan\theta)]+C$

(C) $\frac{(\sec\theta+\tan\theta)}{3}[2+\tan\theta(\sec\theta+\tan\theta)]+C$

(D) $\frac{3(\sec\theta+\tan\theta)}{2}[2+\tan\theta(\sec\theta+\tan\theta)]+C$

MII014

15. $\int\left[\sin\alpha\sin(x-\alpha)+\sin^2\left(\frac{x}{2}-\alpha\right)\right]dx$ equals

(A) $\frac{1}{2}(x+\sin x)+C$

(B) $\frac{1}{2}(x^2-\sin x)+C$

(C) $\frac{1}{2}(x-\sin x)+C$

(D) $\frac{1}{2}(x-\cos x)+C$

MII015

16. $\int\frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$ equals

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(A) $\frac{2}{\alpha+\beta}\sqrt{\frac{x+\beta}{x+\alpha}}+C$

(B) $\frac{-2}{\alpha-\beta}\sqrt{\frac{x-\beta}{x-\alpha}}+C$

(C) $\frac{-2}{\alpha-\beta}\sqrt{\frac{x+\beta}{x+\alpha}}+C$

(D) $\frac{2}{\alpha+\beta}\sqrt{\frac{x-\beta}{x+\alpha}}+C$

MII016

17. $\int\sin^{-1}\sqrt{\frac{x}{a+x}}dx$ equals

(A) $(a+x)\arctan\sqrt{\frac{x}{a}}-\sqrt{ax}+C$

(B) $(a+x)\arctan\sqrt{\frac{a}{x^2}}-\sqrt{ax}+C$

(C) $(a+x^2)\arctan\sqrt{\frac{x^3}{a}}+\sqrt{ax}+C$

(D) $(a+x)\arctan\sqrt{x}+\sqrt{\frac{a}{x}}+C$

MII017

18. $\int\frac{4x^5-7x^4+8x^3-2x^2+4x-7}{x^2(x^2+1)^2}dx$ equals

(A) $4\ell nx-\frac{7}{x}-6\tan^{-1}(x)-\frac{6x}{1-x^2}+C$

(B) $2\ell nx+\frac{1}{x}+6\tan^{-1}(x^2)+\frac{6x}{1-x^2}+C$

(C) $\ell nx-\frac{1}{x}+\tan^{-1}(x)+\frac{x}{1-x^2}+C$

(D) $4\ell nx+\frac{7}{x}+6\tan^{-1}(x)+\frac{6x}{1+x^2}+C$

MII018

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19. $\int \frac{3x^2+1}{(x^2-1)^3}dx$ equals (where K is constant of integration)

(A) $\frac{x}{(x^2-1)^2}+K$

(B) $K-\frac{x}{(x^2+1)^2}$

(C) $K-\frac{x}{(x^2-1)^2}$

(D) $K-\frac{x^2}{(x+1)^2}$

MII019

20. $\int \frac{x^2+3}{x^6(x^2+1)}dx$ equals

(A) $C-\frac{2}{x}+\frac{2}{3x^3}-\frac{3}{5x^5}-2\tan^{-1}x$

(B) $C-\frac{2}{x^2}+\frac{2}{3x^3}-\frac{3}{5x^5}-2\tan^{-1}x$

(C) $C-\frac{2}{x}+\frac{2}{3x^2}-\frac{3}{5x^4}-2\tan^{-1}x$

(D) $C-\frac{2}{x^2}+\frac{2}{3x^4}+\frac{3}{5x^6}+2\tan^{-1}x$

MII020

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. $\int \frac{5x^4+4x^5}{(x^5+x+1)^2}dx$ equals

(A) $C-\frac{x+1}{x^5+x+1}$ (B) $C+\frac{x}{x^5+x+1}$ (C) $C+\frac{x^5}{x^5+x+1}$ (D) $C-\frac{x+1}{x^5-x+1}$

MII041

2. If $I_n = \int (\sin x)^n dx\, n \in N$, then $5I_4 - 6I_6$ is equal to -

(A) $\sin x.(\cos x)^5 + C$ (B) $\cos x.(\sin x)^5 + C$

(C) $\frac{\sin 2x}{8}[\cos^2 2x+1-2\cos 2x]+C$ (D) $\frac{\sin 2x}{8}[\cos^2 2x+1+2\cos 2x]+C$

MII042

3. Suppose $J = \int \frac{\sin^2 x+\sin x}{1+\sin x+\cos x}dx$ and $K = \int \frac{\cos^2 x+\cos x}{1+\sin x+\cos x}dx$. If C is an arbitrary constant of integration then which of the following is/are correct ?

(A) $J = \frac{1}{2}(x - \sin x + \cos x) + C$ (B) $J = K - (\sin x + \cos x) + C$

(C) $J = x - K + C$ (D) $K = \frac{1}{2}(x - \sin x + \cos x) + C$

MII043

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4. If $f\left(\frac{1-x}{1+x}\right)=x$ and $g(x)=\int f(x)dx$ then

(A) $g(x)$ is continuous in domain (B) $g(x)$ is discontinuous at two points in its domain

(C) $\lim_{x\to\infty} g'(x)=-1$ (D) $\int g(x)dx=-\frac{x^2}{2}+(2x+1)\ell n\left(\frac{1+x}{e}\right)+C$

MII044

5. The value of $\int \frac{\ell n\left(\frac{x-1}{x+1}\right)}{x^2-1}dx$ is equal to

(A) $\frac{1}{2}\ell n^2\frac{x-1}{x+1}+C$ (B) $\frac{1}{4}\ell n^2\frac{x-1}{x+1}+C$ (C) $\frac{1}{2}\ell n^2\frac{x+1}{x-1}+C$ (D) $\frac{1}{4}\ell n^2\frac{x+1}{x-1}+C$

MII045

6. If $\int \frac{(x-1)\,dx}{x^2\sqrt{2x^2-2x+1}} = \frac{\sqrt{f(x)}}{g(x)} + C$, where $f(x)$ is a quadratic expression and $g(x)$ is a monic linear expression.

(A) $f(x) = 2x^2-2x+1$ (B) $g(x) = x+1$

(C) $g(x) = x$ (D) $f(x) = 2x^2-2x$

MII046

7. If $\int \frac{4e^x+6e^{-x}}{9e^x-4e^{-x}}dx = Ax + B\,\ell n\,|9e^{2x}-4| + C$, then

(A) $A + 18B = 16$ (B) $18B - A = 19$ (C) $A - 18B = 17$ (D) $A + 18B = 32$

MII047

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8. $\int \frac{x+x^{2/3}+x^{1/6}}{x(1+x^{1/3})}dx = px^{2/3}+q\tan^{-1}(x^{1/6})+c$, then :-

(A) $p < q$ (B) $p = q$ (C) $pq = 9$ (D) $p > q$

MII048

MATRIX MATCH TYPE QUESTION

9.

	List-I		List-II
(P)	$\int \frac{x^2(x^6+x^5-1)dx}{(2x^6+3x^5+2)^2}$	(1)	$-\frac{1}{3}\frac{1}{\left(x^3-x+1\right)}+C$
(Q)	$\int \frac{(x^5+x^4+x^2)dx}{\sqrt{4x^7+5x^6+10x^4}}$	(2)	$\frac{1}{2}(1+x^{-2}+x^{-5})^{-2}+C$
(R)	$\int \frac{(2x^{12}+5x^9)dx}{(x^5+x^3+1)^3}$	(3)	$\frac{-1}{6}(2x^3+3x^2+2x^{-3})^{-1}+C$
(S)	$\int \frac{x^2-\frac{1}{3}}{(x^3-x+1)^2}dx$	(4)	$x\left(\frac{x^3}{25}+\frac{x^2}{20}+\frac{1}{10}\right)^{\frac{1}{2}}+C$

(where C is the constant of integration.)

(A) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 3; S $\rightarrow$ 1 (B) P $\rightarrow$ 3; Q $\rightarrow$ 4; R $\rightarrow$ 2; S $\rightarrow$ 1

(C) P $\rightarrow$ 1; Q $\rightarrow$ 3; R $\rightarrow$ 2; S $\rightarrow$ 4 (D) P $\rightarrow$ 3; Q $\rightarrow$ 1; R $\rightarrow$ 2; S $\rightarrow$ 4

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MII051

MATCHING LIST TYPE QUESTION

10.

	Column-I		Column-II
(A)	Let $f(x) = \int x^{\sin x}\left(1+x\cos x\cdot\ln x+\sin x\right)dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ then the value of $f(\pi)$ is	(P)	rational
(B)	Let $g(x) = \int \frac{1+2\cos x}{(\cos x+2)^2}dx$ and $g(0) = 0$ then the value of $g\left(\frac{\pi}{2}\right)$ is	(Q)	irrational
(C)	Let $k(x) = \int \frac{(x^2+1)dx}{\sqrt[3]{x^3+3x+6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$ then the value of $k(-2)$ is	(R)	integral
(D)	If $\int \frac{\cos x-\sin x+1-x}{e^x+\sin x+x}dx = \ell n\left(f(x)\right)+g(x)+C$ (where C is the constant of integration and $f(x)$ is positive), then $f(0)+g(0)$ is	(S)	prime

MII052

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EXERCISE (S)

1. Let $\int \frac{f'(x)g(x)-g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}}dx=\sqrt{m}\tan^{-1}\left(\sqrt{\frac{f(x)-g(x)}{ng(x)}}\right)+C$,

where $m, n \in N$ and 'C' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$.

MII122

2. A function g defined for all positive real numbers, satisfies $g'(x^2) = x^3$ for all $x > 0$ and $g(1) = 1$. Compute $g(4)$.

MII123

3. Suppose $\int \frac{1-7\cos^2 x}{\sin^7 x\cos^2 x}dx=\frac{g(x)}{\sin^7 x}+C$, where C is arbitrary constant of integration. Then find the value of g'(0) + g"$\left(\frac{\pi}{4}\right)$

MII124

4. If $\int \frac{\cos^5 x}{\sin x}dx=\ell\log|\sin x|+m\sin^4 x+n\sin^2 x+C$, (where C is constant of integration), then $\frac{\ell-n}{m}$ is

MII125

5. If the value $\int \frac{1-(\cot x)^{2008}}{\tan x+(\cot x)^{2009}}dx=\frac{1}{k}\ell n|\sin^k x+\cos^k x|+C$, then find k.

MII053

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6. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$

MII054

7. If the integral $\int \frac{5\tan x}{\tan x-2}dx = x + a\,\ell n|\sin x - 2\cos x| + k$ then a is equal to :

MII055

8. If $\int \frac{x^4-2x^2-3}{x^4\sqrt{x(1+x^{-2})^2+1}}dx=2(a+bx+cx^{-1}+dx^{-3})^{\frac{1}{2}}+K$, where K is the integration constant, then the value of $a + b + c + d$ is

MII056

9. Let F is anti-derivative of f defined by $f(x) = 5(x^4 - 1)$ where $F(1) = 20$, then value of $F(2)$ is

MII057

10. $\int \frac{dx}{16+x^2}=f(x)$ where $f(4)=\frac{\pi}{16}$ and ($\pi=3.14$), then value of $f(2)+f\left(\frac{4}{3}\right)$ is –

MII058

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EXERCISE - JEE (Main) PYQ

1. For $x^2 \neq n\pi + 1, n \in N$ (the set of natural numbers), the integral

$\int x\sqrt{\frac{2\sin\left(x^2-1\right)-\sin 2\left(x^2-1\right)}{2\sin\left(x^2-1\right)+\sin 2\left(x^2-1\right)}}dx$ is equal to :

(where c is a constant of integration) **[JEE (Main) 2019]**

(1) $\log_e\left|\sec\left(\frac{x^2-1}{2}\right)\right|+c$ (2) $\log_e\left|\frac{1}{2}\sec^2\left(x^2-1\right)\right|+c$

(3) $\frac{1}{2}\log_e\left|\sec^2\left(\frac{x^2-1}{2}\right)\right|+c$ (4) $\frac{1}{2}\log_e\left|\sec(x^2-1)\right|+c$

MII032

2. If $\int\frac{\sqrt{1-x^2}}{x^4}dx = A(x)\left(\sqrt{1-x^2}\right)^m + C$, for a suitable chosen integer m and a function $A(x)$, where C is a constant of integration then $(A(x))^m$ equals : **[JEE (Main) 2019]**

(1) $\frac{-1}{3x^3}$ (2) $\frac{-1}{27x^9}$ (3) $\frac{1}{9x^4}$ (4) $\frac{1}{27x^6}$

MII033

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3. The integral $\int\cos(\log_e x)dx$ is equal to : (where C is a constant of integration) **[JEE (Main) 2019]**

(1) $\frac{x}{2}[\sin(\log_e x)-\cos(\log_e x)]+C$ (2) $\frac{x}{2}[\cos(\log_e x)+\sin(\log_e x)]+C$

(3) $x[\cos(\log_e x)+\sin(\log_e x)]+C$ (4) $x[\cos(\log_e x)-\sin(\log_e x)]+C$

MII034

4. If $\int\frac{\cos x\,dx}{\sin^3 x\left(1+\sin^6 x\right)^{2/3}} = f(x)\left(1+\sin^6 x\right)^{1/\lambda}+c$

where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to **[JEE (Main) 2020]**

(1) –2 (2) $-\frac{9}{8}$ (3) 2 (4) $\frac{9}{8}$

MII035

5. If $\int\frac{d\theta}{\cos^2\theta(\tan 2\theta+\sec 2\theta)} = \lambda\tan\theta + 2\log_e|f(\theta)| + C$ where C is a constant of integration, then the ordered pair $(\lambda, f(\theta))$ is equal to : **[JEE (Main) 2020]**

(1) (–1, 1 + tanθ) (2) (–1, 1 – tanθ) (3) (1, 1 – tanθ) (4) (1, 1 + tanθ)

MII036

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6. The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to :

(where C is a constant of integration) **[JEE (Main) 2020]**

(1) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}}+C$ (2) $-\left(\frac{x-3}{x+4}\right)^{-\frac{1}{7}}+C$ (3) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}}+C$ (4) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{-\frac{13}{7}}+C$

MII037

7. If $\int \frac{\cos x-\sin x}{\sqrt{8-\sin 2x}}dx = a\sin^{-1}\left(\frac{\sin x+\cos x}{b}\right)+c$, where c is a constant of integration, then the ordered pair (a, b) is equal to : **[JEE (Main) 2021]**

(1) (–1, 3) (2) (3, 1) (3) (1, 3) (4) (1, –3)

MII038

8. $\int \frac{\left(x^2+1\right)e^x}{\left(x+1\right)^2}dx = f\left(x\right)e^x+C$, Where C is a constant, then $\frac{d^3f}{dx^3}$ at $x = 1$ is equal to :

[JEE (Main) 2022]

(1) $-\frac{3}{4}$ (2) $\frac{3}{4}$ (3) $-\frac{3}{2}$ (4) $\frac{3}{2}$

MII039

9. For $I(x)=\int \frac{\sec^2 x-2022}{\sin^{2022} x}dx$ if $I\left(\frac{\pi}{4}\right)=2^{1011}$, then **[JEE (Main) 2022]**

(1) $3^{1010}I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (2) $3^{1010}I\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

(3) $3^{1011}I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (4) $3^{1011}I\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

MII040

10. For, $\alpha,\beta,\gamma,\delta\in\mathbb{N}$, if $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$,

Where $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ and C is constant of integration, then $\alpha + 2\beta + 3\gamma - 4\delta$ is equal to:

[JEE (Main) 2023]

(1) 1 (2) –4 (3) –8 (4) 4

MII064

11. Let $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$.

If $f(3)=\frac{1}{2}(\log_e 5-\log_e 6)$, then $f(4)$ is equal to **[JEE (Main) 2023]**

(1) $\frac{1}{2}(\log_e 17-\log_e 19)$ (2) $\log_e 17-\log_e 18$

(3) $\frac{1}{2}(\log_e 19-\log_e 17)$ (4) $\log_e 19-\log_e 20$

MII065

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12. Let $I(x) = \int \sqrt{\frac{x+7}{x}}\,dx$ and $I(9) = 12 + 7\log_e 7$. If $I(1) = \alpha + 7\log_e\left(1+2\sqrt{2}\right)$, then α^4 is equal to_____. **[JEE (Main) 2023]**

MII066

13. If $\int \sqrt{\sec 2x - 1}\,dx = \alpha \log_e \left|\cos 2x + \beta + \sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|$ + constant, then $\beta - \alpha$ is equal to ____.

[JEE (Main) 2023]

MII067

14. Let $I(x) = \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{\left(x\tan x + 1\right)^2}dx$. If $I(0) = 0$ the $I\left(\frac{\pi}{4}\right)$ is equal to **[JEE (Main) 2023]**

(1) $\log_e \frac{(\pi+4)^2}{16} - \frac{\pi^2}{4(\pi+4)}$ (2) $\log_e \frac{(\pi+4)^2}{16} + \frac{\pi^2}{4(\pi+4)}$

(3) $\log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$ (4) $\log_e \frac{(\pi+4)^2}{32} + \frac{\pi^2}{4(\pi+4)}$

MII068

15. If $\int \frac{1}{a^2\sin^2 x + b^2\cos^2 x}dx = \frac{1}{12}\tan^{-1}(3\tan x)$ + constant, then the maximum value of $a\sin x + b\cos x$, is : **[JEE (Main) 2024]**

(1) $\sqrt{40}$ (2) $\sqrt{39}$ (3) $\sqrt{42}$ (4) $\sqrt{41}$

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MII126

16. If $\int \frac{1}{\sqrt[5]{(x-1)^4\,(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$, where C is the constant of integration, then the value of $\alpha + \beta + 20AB$ is _____ . **[JEE (Main) 2024]**

MII127

17. Let $I(x) = \int \frac{6}{\sin^2 x(1-\cot x)^2}dx$. If $I(0) = 3$, then $I\left(\frac{\pi}{12}\right)$ is equal to : **[JEE (Main) 2024]**

(1) $\sqrt{3}$ (2) $3\sqrt{3}$ (3) $6\sqrt{3}$ (4) $2\sqrt{3}$

MII128

18. For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, if $y(x) = \int \frac{\operatorname{cosec} x + \sin x}{\operatorname{cosec} x \sec x + \tan x \sin^2 x}dx$ and $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

[JEE (Main) 2024]

(1) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (2) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (3) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (4) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$

MII129

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EXERCISE - JEE (Advanced) PYQ

1. $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$ is equal to - **[IIT JEE 2006]**

(A) $\frac{\sqrt{2x^4-2x^2+1}}{x^2}+c$ (B) $\frac{\sqrt{2x^4-2x^2+1}}{x^3}+c$

(C) $\frac{\sqrt{2x^4-2x^2+1}}{x}+c$ (D) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+c$

MII059

2. Let $f(x)=\frac{x}{(1+x^n)^{1/n}}$ for $n\geq 2$ and $g(x)=\underbrace{(f\circ f\circ ...\circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2}\,g(x)dx$ equals.

(A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}}+K$ (B) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}}+K$ **[IIT JEE 2007]**

(C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}}+K$ (D) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}}+K$

MII060

3. Let $F(x)$ be an indefinite integral of $\sin^2 x$. **[IIT JEE 2007]**

Statement-1 : The function $F(x)$ satisfies $F(x+\pi)=F(x)$ for all real x.

because

Statement-2 : $\sin^2(x+\pi)=\sin^2 x$ for all real x.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

MII061

4. Let $I=\int \frac{e^x}{e^{4x}+e^{2x}+1}dx$, $J=\int \frac{e^{-x}}{e^{-4x}+e^{-2x}+1}dx$. Then, for an arbitrary constant c, the value of $J-I$ equals **[IIT JEE 2008]**

(A) $\frac{1}{2}\ell n\left(\frac{e^{4x}-e^{2x}+1}{e^{4x}+e^{2x}+1}\right)+c$ (B) $\frac{1}{2}\ell n\left(\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right)+c$

(C) $\frac{1}{2}\ell n\left(\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right)+c$ (D) $\frac{1}{2}\ell n\left(\frac{e^{4x}+e^{2x}+1}{e^{4x}-e^{2x}+1}\right)+c$

MII062

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5. The integral $\int \frac{\sec^2 x}{(\sec x+\tan x)^{9/2}}dx$ equals (for some arbitrary constant K)

[JEE (Advanced) 2012]

(A) $-\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}-\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

(B) $\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}-\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

(C) $-\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}+\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

(D) $\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}+\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

MII063

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	B	D	C	B	A	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	C	C	B	A	D	C	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A,C	B,C	B,C	A,C	B,D	A,C	A,B	A,C	B	
Que.	10									
Ans.	A → Q, B → P, C → P,R,S ,D → P,R									

EXERCISE (S)

1. 8 **2.** 13.40 **3.** 5 **4.** 8 **5.** 2010
6. 3 **7.** 2 **8.** 5 **9.** 46
10. 0.19 or 0.20

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1 or 3	2	2	1	1	1	3	2	1	4
Que.	11	12	13	14	15	16	17	18		
Ans.	1	64	1	3	1	7	2	4		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	
Ans.	D	A	D	C	C	

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Important Notes

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